Bandit-Based Material Screening

Why it matters

- Every alloy test costs time, material and money
- Uniform testing wastes budget on weak candidates
- A bandit decides which alloy to test next, trading off exploring and exploiting

Application

Screening candidate steels for high yield strength (MPa) under a fixed test budget

Original problem statement

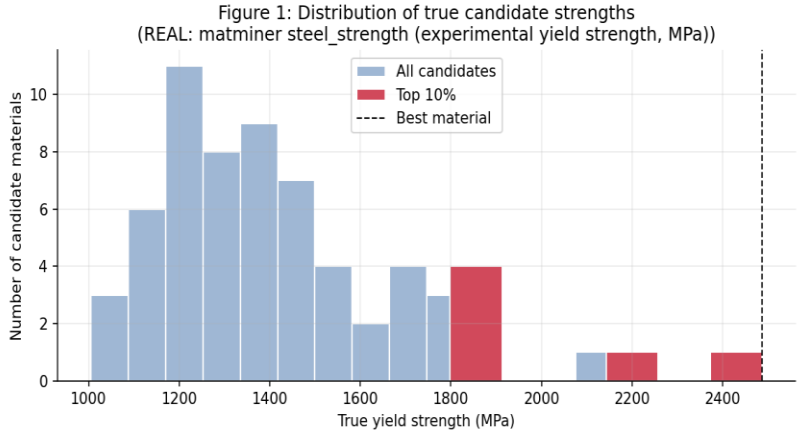
"A multi-armed or contextual bandit that chooses which candidate alloys or compounds to 'test' next from a public dataset (e.g., Matbench)."

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Environment and bandit formulation

Environment	60 candidate steels, hidden true strength, noisy tests
Arm / Action	Each steel is an arm; the action is which steel to test next
State	Stateless; belief summarised by counts and mean estimates
Reward	Observed strength = true mean + Normal noise (SD 100 MPa)
Horizon	300 tests, same for every algorithm
Objective	Maximise reward, minimise regret, find the best steel



Pool: 60 real steels sampled from matminer steel_strength. Reported yield strengths are the hidden true means. Measurement noise is simulated.

Algorithms compared

Random

$P(a) = 1/K$

No learning. Pure exploration. The baseline.

Epsilon-Greedy

$Q \leftarrow Q + (r - Q) / N$

Exploit the best Q; explore at random with probability 0.1.

UCB1

$\text{argmax } Q + c\sqrt{(2 \ln t / N)}$

Optimism: rarely tested alloys get a bonus. c = 1 standard.

Thompson Sampling

sample $\mu \sim N(m, 1/\lambda)$, play argmax

Sample beliefs; explore where uncertainty is high.

Results across 30 seeds

Figure 2: Cumulative observed reward

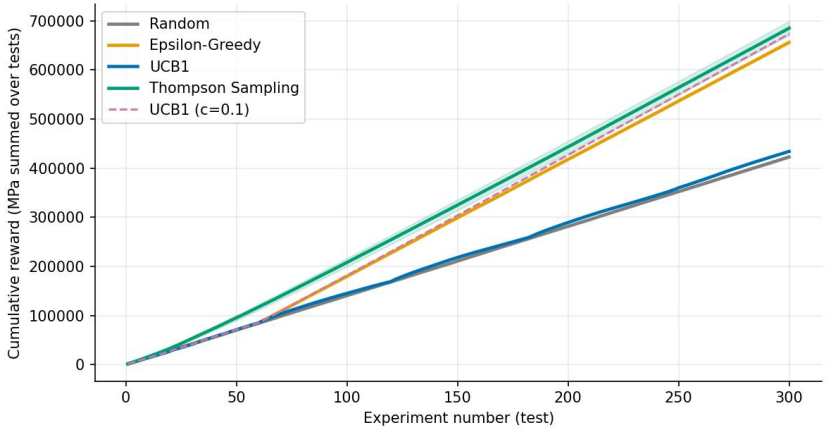
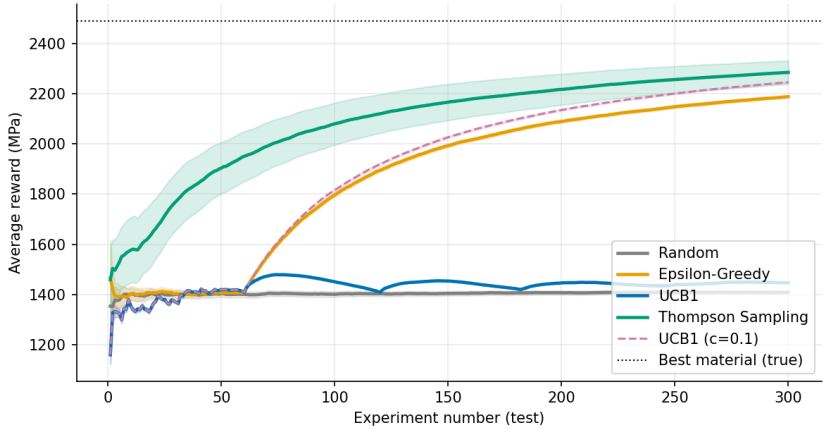


Figure 3: Average observed reward per experiment (learning curve)



Algorithm	Cumulative reward (MPa)	Cumulative regret (MPa)	Best-steel ID
Random	422,827	323,156	100%
Epsilon-Greedy	656,257	89,898	100%
UCB1	434,094	312,026	100%
Thompson Sampling	685,205	60,954	93%

Mean over 30 seeds, 300 tests, real steel pool, simulated noise. Oracle reward would be about 746,190. Plain UCB1 (c = 1) is still exploring at this budget.

Conclusion, limitations, ethics

Conclusion

- Learning methods beat random testing
- Thompson Sampling: highest reward and lowest regret; best steel found in 93% of runs
- UCB1 needs a smaller exploration scale at this budget

Limitations

- Noise is simulated: dataset values treated as true means
- Only 60 of 312 steels, one budget setting
- Arms independent; no composition features used

Ethics

- Never present simulated values as measurements
- Screening aid only; safety-critical use needs validation
- Public data may favour well-studied alloys